So now that we have a better idea of what basis is, we want to understand better how they work. One big question we alluded to was: do two bases for the same space have to have the same number of vectors?

Mentioned how $\mathbb{R}^3$ had natural basis $\{e_1, e_2, e_3\}$

We then wondered if we had a spanning set for $\mathbb{R}^3$ with say five vectors if that was assured to be redundant. Felt like it ought to be, if we could span it with three why would we need five elsewhere? But ok we have to prove that.

As alluded, for $\mathbb{R}^n$, we have the <u>standard basis:</u> $\{e_1, e_2, \ldots, e_n\}$ where

$$e_1 = (1, 0, 0, \ldots, 0, 0)$$
$$e_2 = (0, 1, 0, \ldots, 0, 0)$$
$$\vdots$$
$$e_n = (0, 0, 0, \ldots, 0, 1)$$

That is, e_k has 1 in the k-th position, 0s elsewhere for each $1 \leq k \leq n$. We called these "standard vectors" before, just didn't use the term basis then as we hadn't yet defined that term. So this kind of formalizes that idea.

This is most common basis used for $\mathbb{R}^n$ for several reasons (hence why it is referred to as the standard basis). Why? Well say in $\mathbb{R}^3$ we took another basis:

$$\mathcal{B} = \{ (1, 2, 3), (4, 5, 20), (10, 11, 99) \}$$

(this is arbitrary basis nothing inherently special).

In $\mathbb{R}^3$ say we take vector joining the origin to point (3, 7, 15)

How do we represent that vector? In standard basis very straightforward, $3e_1 + 7e_2 + 15e_3$, i.e. (3, 7, 15), coordinates very apparent.

But what would its coordinates in $\mathcal{B}$ be? Well have to solve

$$c_1(1, 2, 3) + c_2(4, 5, 20) + c_3(10, 11, 99) = (3, 7, 15)$$

So have to solve that system which takes some work, certainly doable but clearly standard basis much easier.

For any x we use this notation:

$[x]_{\mathcal{B}}$: coordinates of x in $\mathcal{B}$

x: coordinates of x in standard basis unless otherwise specified

Natural question is if we have the representation of a vector in one of these forms, how do we switch to other?

E.g. using the $\mathcal{B}$ mentioned earlier, if we have

$$[x]_{\mathcal{B}} = (5, 6, 13)$$

then what is x (i.e. coordinates in standard basis)?

Say we wanted to draw this vector in same way as earlier, connecting origin to appropriate point. But having its coordinates in this strange basis doesn't really help with that does it? We really need its coordinates in standard basis to do that conveniently.

Hence the motivation for this. But how to switch that?

Make matrix whose columns are vectors in $\mathcal{B}$ using same prescribed order as in coordinate system for $\mathcal{B}$.

$$\begin{bmatrix} 1 & 4 & 10 \\ 2 & 5 & 11 \\ 3 & 20 & 99 \end{bmatrix} \begin{bmatrix} 5 \\ 6 \\ 13 \end{bmatrix} = \begin{bmatrix} 159 \\ 183 \\ 1422 \end{bmatrix}$$

representation in standard basis

Why? Well this matrix multiplication is just 5 times first column (first vector), 6 times second vector, etc. And that's exactly what coordinates mean so that's why this works.

Don't really need matrices to do this, but it's a convenient way to write it.

4.4.3

$$\mathcal{B} = \left\{ \begin{bmatrix} 1 \\ -4 \\ 3 \end{bmatrix}, \begin{bmatrix} 5 \\ 2 \\ 2 \end{bmatrix}, \begin{bmatrix} 4 \\ -7 \\ 0 \end{bmatrix} \right\} , \quad [x]_{\mathcal{B}} = \begin{bmatrix} 3 \\ 0 \\ -1 \end{bmatrix}$$

Again to find x (representation, in standard basis):

$$\begin{bmatrix} 1 & 5 & 4 \\ -4 & 2 & -7 \\ 3 & 2 & 0 \end{bmatrix} \begin{bmatrix} 3 \\ 0 \\ -1 \end{bmatrix} = \begin{bmatrix} -1 \\ -5 \\ 9 \end{bmatrix}$$

4.4.7

$$\underbrace{b_1 = \begin{bmatrix} 1 \\ -1 \\ 3 \end{bmatrix}, \quad b_2 = \begin{bmatrix} -3 \\ 4 \\ 9 \end{bmatrix}, \quad b_3 = \begin{bmatrix} 2 \\ 2 \\ 4 \end{bmatrix}}_{\mathcal{B}}$$

$$x = \begin{bmatrix} 8 \\ -9 \\ 6 \end{bmatrix}$$

Switch x into $\mathcal{B}$-coordinates?

Basically the x we found in prior question is now known while the vector we multiply the matrix by becomes the unknown. So must solve

$$\begin{bmatrix} 1 & -3 & 2 \\ -1 & 4 & -2 \\ 3 & 9 & 4 \end{bmatrix} \begin{bmatrix} c_1 \\ c_2 \\ c_3 \end{bmatrix} = \begin{bmatrix} 8 \\ -9 \\ 6 \end{bmatrix}$$

Use augmented matrix (can use inverse too)

$$\begin{bmatrix} 1 & -3 & 2 & 8 \\ -1 & 4 & -2 & -9 \\ 3 & 9 & 4 & 6 \end{bmatrix}$$

Replace R2 with R2 + R1 and R3 with R3 - 3R1 to get

$$\sim \begin{bmatrix} 1 & -3 & 2 & 8 \\ 0 & 1 & 0 & -1 \\ 0 & 18 & -2 & -18 \end{bmatrix}$$

Replace R3 with R3 - 18R2:

$$\sim \begin{bmatrix} 1 & -3 & 2 & 8 \\ 0 & 1 & 0 & -1 \\ 0 & 0 & -2 & 0 \end{bmatrix}$$

From this triangular form of coefficient matrix we get:

$$c_3 = 0, c_2 = -1, c_1 = 5$$

So representation is:

$$\begin{bmatrix} 5 \\ -1 \\ 0 \end{bmatrix}_B$$

4.5 Dimension

Very important result: If a real vector space V has a finite basis with n elements, then every basis for V has n elements.

Another important result, the one which mainly proves the above is that in such a space, any set of $> n$ vectors is linearly dependent.

Why is latter true? Say V has a basis $\{b_1, b_2, b_3, b_4, b_5\}$

Then consider set of vectors in V: $\{w_1, w_2, \ldots, w_7\}$

Basis with five elements, and we take a set of seven. We will prove this set is linearly dependent to verify our statement. Yes, taking a specific case to make the notation easier, but we can generalize this once we show this case without much trouble.

To check linear dependence of the w_k's need to assess

$$c_1 w_1 + c_2 w_2 + \ldots + c_7 w_7 = 0$$

Does this have a nontrivial solution for the c_i's is point.
Since the b_j's are a basis for the space, we have

$$\begin{aligned} w_1 &= r_{11} b_1 + r_{12} b_2 + \ldots + r_{15} b_5 \\ w_2 &= r_{21} b_1 + r_{22} b_2 + \ldots + r_{25} b_5 \\ &\vdots \\ w_7 &= r_{71} b_1 + r_{72} b_2 + \ldots + r_{75} b_5 \end{aligned}$$

That is, every vector in V can be written as a linear combination of the b_j's since they span the space.

Sub these expressions for the w_k's into the earlier linear combination, to find that equivalent to

$$c_1[r_{11}b_1 + \ldots + r_{15}b_5] + c_2[r_{21}b_1 + \ldots + r_{25}b_5]$$

$$+ \ldots + c_7[r_{71}b_1 + \ldots + r_{75}b_5] = 0$$

Collecting the b_j's this can be rearranged to:

$$[c_1 r_{11} + \ldots + c_7 r_{71}]\, b_1 + \ldots + [c_1 r_{15} + \ldots + c_7 r_{75}]\, b_5 = 0$$

For this to hold the coefficient of each b_j must be 0 since the b_j's are linearly independent (they are a basis). Thus the original equation we were trying to solve is equivalent to the matrix equation:

$$\begin{bmatrix} r_{11} & r_{21} & \cdots & r_{71} \\ r_{12} & r_{22} & \cdots & r_{72} \\ \vdots & \vdots & & \vdots \\ r_{15} & r_{25} & \cdots & r_{75} \end{bmatrix} \begin{bmatrix} c_1 \\ c_2 \\ \vdots \\ c_7 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ \vdots \\ 0 \end{bmatrix}$$

This again is equivalent to the linear dependence equation we were checking earlier. So again do c_i's have a nontrivial solution? Well coefficient matrix of this homogeneous equation has more columns than rows guaranteeing a nonpivot column so infinite solutions.

Thus c_i's have nontrivial solution for sure guaranteeing the w_k's are linearly dependent, as desired.

And again five basis elements with seven other vectors isn't important, key is more of latter type will result in matrix with more columns, guaranteeing homogeneous system with nontrivial solution, so linearly dependent.

So if V has basis with n elements, every basis has n vectors. Can't have more than n as above result shows a larger set will be linearly dependent and thus not a basis.

And smaller set with say $k < n$ vectors can't be basis as then again above result would show set with $> k$ vectors would then be linearly dependent, contradicting that the original n element set was actually a basis.

So if V has finite basis of n elements, we call n the <u>dimension</u> of V, denoted $\dim V$.

This is well defined since size of basis is fixed, without that concept of dimension wouldn't really make sense.

Some vector spaces have finite basis, naturally are called <u>finite dimensional</u>. Those without finite basis are naturally called <u>infinite dimensional</u>.

Dimension helps make many things clearer. When we were say thinking about a linear transformation from $\mathbb{R}^3$ to $\mathbb{R}^2$, we found via matrix ideas that such a transformation could never be one-to-one. But could you argue that it can't be one-to-one because the domain is "bigger" than the codomain?

That's rather dicey as these are infinite sets so concept of "bigger" is much more nebulous. But if we instead said it can't be one-to-one because the domain has a higher dimension than the codomain (3 vs. 2), that concretely makes much more sense as then you are comparing two finite numbers.

And indeed we can prove it that way, as say the three standard basis elements in $\mathbb{R}^3$ must be mapped to a linearly dependent set in $\mathbb{R}^2$ since three vectors in a space of dimension two are always linearly dependent. And via this we can map a nonzero vector to zero vector.

Another way this helps is subspace always has dimension $\leq$ that of full space.

An interesting question is if a subspace has equal dimension to that of full space must it actually equal the full space? (Answer is yes).

Another thing: If $\dim V = n$, then any linearly independent set of n vectors in V is a basis for V.

Why? Say you had such a linearly independent set and it wasn't a basis. Then some v in V is missing from the span. Put v in the linearly independent set.

Since v wasn't in the span, new set still linearly independent. But then a space of dimension n has a linearly independent set of $n + 1$ vectors which we said is not possible. So original must have been basis.

For an $m \times n$ real matrix A what are dimensions of Nul A, Col A, Row A?

dim (Nul A): number of nonpivot columns in A
dim (Col A): number of pivot columns in A
dim (Row A): number of pivots in A so same as dim (Col A)

Other day had nonsquare matrix which had same size basis for column and row space.

Seemed surprising since they both lie in different spaces and had different number of starting vectors.

Maybe that was just a coincidence?

As this shows it wasn't! Row and column space always have same dimension, regardless of size of matrix.

That's used enough that we call that dimension the <u>rank</u> of the matrix.

And the dimension of the null space is the <u>nullity</u> of matrix.

Since former equal to number of pivot columns and latter to number of nonpivot columns, that gives us:

<u>Rank-Nullity Theorem</u>

For any real $m \times n$ matrix A we have:

$$\text{rank}(A) + \text{nullity}(A) = n$$

This is a very useful fact. One style of situation where we might use it is when thinking of some generic situation where we want say the size of the range of a linear transformation, i.e. its dimension.

Assuming that linear transformation is a matrix transformation (again not always assured, that requires domain and codomain to be of finite dimension), asking for the "size" of the range, means we want the dimension of the column space of the associated matrix, i.e. its rank.

Again, if we knew the matrix, then just via REF we can find that. But if situation is more generic, not given that, given more speculative style information (e.g. we are told our desired transformation is the composition of two transformations having certain ranks), we might find it easier to assess the nullity of the matrix as checking the size of what vectors get sent to 0 is often an easier thing to do.

Another thing rank is used is that for a square matrix, n x n, to be invertible must have trivial null space (Invertible Matrix Theorem). So nullity of such matrix is always 0, thus rank is n. So invertible n x n matrix always has rank n, i.e "full rank".

Rank is useful beyond this. Say you have a 10 x 10 matrix, whose rank is lower than 10. It's definitely noninvertible then, but we might handle it very differently if rank is say 8 compared to 5.

Just like determinant, rank is used for more than just figuring out invertibility, in fact we use it in nonsquare matrices too unlike determinant.